

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage.
Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment.
Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments.
Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

Course Code: CSA1521

Course Name: Cloud Computing & Big

Data Analytics

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CO2-AT3- Application.

Based Questions.

Q1) AUTO SCALING AND LOAD BALANCING:

A company facing unpredictable traffic during seasonal scales can use cloud auto-scaling and load balancing.

- >> Auto Scaling automatically increases the number of servers when customer traffic increases.
- >> When traffic decreases, unnecessary servers are removed reducing resource usage.
- >> Load balancing distributes incoming requests among multiple servers so that all one server does not become overloaded. This reduce downtime, improves response time, and provides a better shopping experiences.
- >> Monitoring tools continuously tracks CPU usage, memory trigger scaling when required.

Q2) CLOUD STORAGE, DURABILITY AND DATA REDUNDANCY:

Cloud storage is suitable for storing large amount of backup data securely.

⇒ Cloud provides store data using multiple copies, reducing the risk of losing information when hardware fails.

⇒ Data can be replicated across different storage locations or region for better redundancy.

⇒ Encryption protects backup data from unauthorized access during storage and transmission.

⇒ Access control allows only authorized users or administrators to access sensitive backup files.

⇒ Cloud storage can be more cost-effective because organizations can use different storage classes according to how frequently backups are accessed.

Disaster Recovery Example:

If a company's main database is accidentally deleted or become unavailable, its cloud backup can be used to restore the database and continue business operations.

Q3)

CDN and Edge Servers:

Mobile applications serving users globally can use Content Delivery Networks (CDNs) and edge servers to improve performance.

- ★ A CDN stores frequently requested content at locations closer to users.

- ★ Users receive content from a nearby CDN location instead of always contacting the main cloud server.

- ★ This reduces network latency and improves applications response time.

- ★ Edge servers process certain requests closer to users, providing faster responses across different regions.

- ★ Cloud providers can deploy applications across multiple regions, improving global availability if one region experiences problems.

Real World Example:

In an online gaming application, game updates

and static files can be delivered through nearby edge locations, reducing delay for players in different countries.

Q4) Cloud Compliance, Security AND DATA SOVEREIGNTY:

~~A~~ financial institution needs strong security and compliance controls when using cloud services.

*→ Logging records user activities, system events, and access attempts for auditing purposes.

*→ Monitoring helps detect suspicious activities, security incidents, and unusual system behaviour.

*→ Encryption protects sensitive financial information from unauthorized access.

*→ Access control and IAM ensure that only authorized employees can access specific financial resources.

*→ Data sovereignty can be maintained by storing sensitive data in cloud regions permitted by the organization's regulations.

Example:.

For an online banking system, customer transaction data can be encrypted and stored in an approved geographical region, while access logs are maintained for regulatory audits.

Q5) CONTAINERS, VIRTUAL MACHINES AND KUBERNETES:

Containers provide a lightweight method for deploying applications compared with traditional virtual machines.

➡ Containers share the host operating system kernel, so they generally require fewer resources than separate virtual machines.

➡ Virtual Machines include a complete guest operating system, which increases resource usage.

➡ Containers can start quickly, making them useful for rapidly deploying applications.

➡ Kubernetes manages containers, automatically handling deployment, service management, and scaling.

➡ Containers provide portability, allowing the same applications to run across different cloud environments with fewer changes.

Microservices Example:

An e-commerce applications can use separate containers for user login, product catalog, payments, and order management. Kubernetes can deploy and scale each service independently.
